# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Manufactured precision composite components for aerospace and renewable energy sectors by setting up and operating **3-axis and 5-axis CNC machines** to exact technical drawings and specifications.
- Loaded and unloaded materials into CNC machines, ensuring proper **machine alignment and calibration** for accurate operation and selecting appropriate tooling for each job.
- Created and modified **CNC programmes and simulations** for specific tasks, monitoring machine performance and providing detailed feedback to key stakeholders.
- Inspected finished parts using **precision measuring tools** to ensure they met required tolerances and specifications, documenting and reporting any deviations.
- Performed basic maintenance on CNC machines, including **cleaning, lubrication, and tool replacement**, ensuring machines remained in optimal working condition to minimize downtime.
- Collaborated closely with engineers and production teams to optimize production efficiency, completing ongoing training and seeking mentoring to support learning needs in **advanced composite manufacturing**.
- Adhered to all safety protocols and procedures, ensuring machines and tools were properly maintained to avoid accidents and complying with company policies and environmental regulations.

## SKILLS

**CNC Machining** – 3-axis CNC, 5-axis CNC, Machine Setup, Machine Operation, Tooling, Calibration, Programming, Simulations

**Quality Control** – Technical Drawings, Specifications, Tolerances, Precision Measuring Tools, GD&T;, Inspection, Documentation

**Manufacturing Processes** – Composite Manufacturing Methods, Material Loading/Unloading, Machine Maintenance, Lubrication, Tool Replacement

**Software & Tools** – CAD/CAM, SolidWorks, Hypermill, Microsoft Office, WORD, EXCEL

**Professional Skills** – Problem-Solving, Machine Troubleshooting, Team Collaboration, Communication, Attention to Detail, Adaptability

## PROJECTS

### Precision Component Machining (HCL Technologies)

- Executed precise machining operations on a range of enterprise components using **3-axis CNC machinery**, adhering strictly to complex technical drawings and material specifications.
- Managed the full lifecycle of component production, from initial **tool selection and setup** to quality inspection using calipers and micrometers, ensuring adherence to tight tolerances.
- Maintained operational readiness of CNC equipment through scheduled **preventative maintenance and basic repairs**, contributing to an overall reduction in unscheduled downtime by approximately 15%.

### Advanced Composite Part Development (HCL Technologies)

- Operated and set up **5-axis CNC machines** for intricate composite part fabrication, directly translating engineering models into physical components.
- Performed detailed quality checks on finished parts against **geometric dimensioning and tolerancing (GD&T;)** standards, documenting any deviations for engineering review.
- Contributed to process improvement initiatives by providing feedback on machine performance and tooling wear, supporting the team's goal of increasing throughput by 10%.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)